

Transcutaneous External Pacing

Guideline Title: Transcutaneous External Pacing

Guideline Number: GUID-3203-PR022

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Revision dates: 6/8/2021	Developed by: Air Care Team
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CCCXXI. **PURPOSE:** To improve perfusion and oxygenation due to symptomatic bradycardia

CCCXXII. **DEFINITIONS:**

- A. TCP – Transcutaneous Pacing
- B. SBP – Systolic Blood Pressure

CCCXXIII. **DEPARTMENT GUIDELINE:**

A. **EQUIPMENT:**

- i. Cardiac monitor with pacing capabilities
- ii. External conductive hands-free pad electrodes and cable

B. **INDICATIONS:**

- i. Symptomatic bradycardia unresponsive to medication administration.
 - 1. Pulse < 50 with hypotension (SBP < 90)
 - 2. Symptoms: altered mental status, chest pain, and/or dyspnea
- ii. Second or third-degree heart block

C. **CONTRAINDICATIONS:** There are no contraindications for the truly symptomatic patient

D. **PROCEDURE:**

- i. Assess the need for TCP. Confirm brady-dysrhythmia.
- ii. Consider sedation or analgesia per *GUID-3203-G005 Analgesia and Sedation Management*.
- iii. Turn external pacemaker on.
- iv. Attach standard 4 lead ECG cable.
- v. Place pacing pads on patient's bare chest:
 - 1. Anterior / Posterior (preferred): place negative electrode on left anterior chest, halfway between the xiphoid process and the left nipple. This corresponds with the V2-V3 ECG

position. Place the positive electrode on left posterior chest beneath the scapula and lateral to the spine.

2. Anterior / Anterior: place the negative electrode on the left chest, mid-axilla over the fourth intercostal space. Place positive electrode on anterior right chest, sub-clavicular area.
- vi. Select Lead I, II, III to provide the largest amplitude QRS complex.
- vii. Push the “Pacer” or locate pacer function.
- viii. Select DEMAND mode.
- ix. Select desired ‘RATE’ (typically 10-20 beats above the patient’s intrinsic rate, or 60 if no rate exists)
- x. Observe for a pacing stimulus marker for each “R” wave. If pacing marker does not exist with each “R” wave, increase the ECG size gain until the sensing marker appears or change the ECG lead.
- xi. When sensing is confirmed, increase mA OUTPUT by 10 mA increments, until electrical capture is determined (evidenced by a wide QRS and tall, broad T wave).
- xii. Determine OPTIMUM THRESHOLD. Maintain stimulation at or 10% above lowest mA value that maintains capture.
- xiii. Mechanical or ventricular capture is evidenced by signs of improving cardiac output. Palpate for a femoral, right brachial, or radial artery pulse.
- xiv. Assess every 5 minutes: blood pressure, pulse, respirations, SpO2, level of consciousness, and ECG rhythm.
- xv. The non-invasive pacing may produce skeletal muscle contractions. Secure all tubing, cables, and other equipment to prevent their displacement

CCCXXIV. **DOCUMENTATION:** EMS Charts

CCCXXV. **REFERENCES:**

- A. American Heart Association, 2015.
- B. Zoll – X Series for EMS www.zoll.com